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DESCRIPTION

This document relates to Signa Horizon Release 8.X Operator Workspace Subsystem.

1- INTRODUCTION

The purpose of this document is to outline the details for the Operator Workspace subsystem. The Operator Workspace has two main components. The Operator Workspace Table, and the Operator Workspace Cabinet. See Illustration L1. The Table has a monitor that is the user interface to the Host computer, and an LCD panel that is the user interface to the IBM Compatible PC.

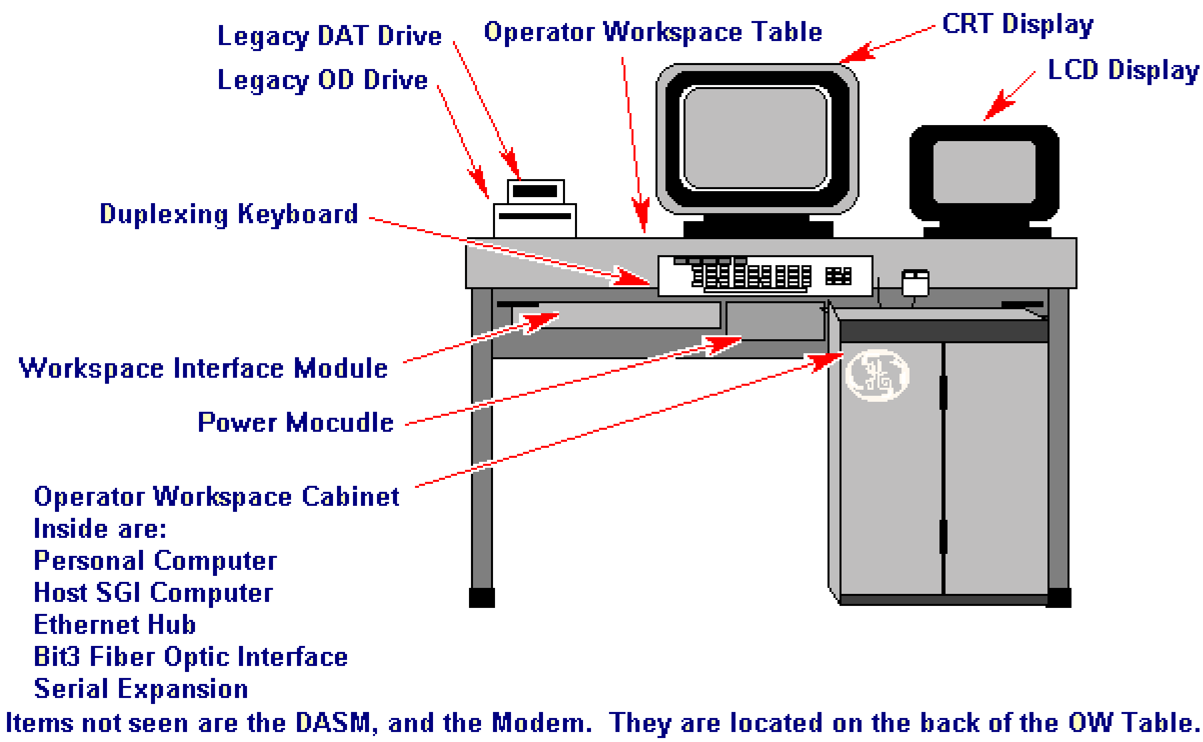


ILLUSTRATION L1
OPERATOR WORK SPACE WITH INDIGO COMPUTER

The host computer is based on a Silicon Graphics Inc. (SGI) Indigo II^{TM} workstation board set. Included is a modified back plane to allow more add-on boards than the standard SGI workstation. See Illustration L5500A for a simplified block diagram of the SGI Host Computer Chassis.

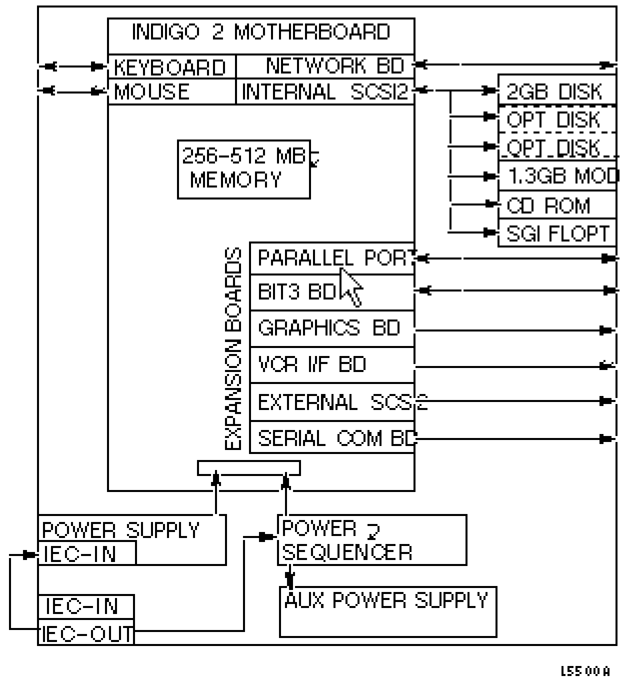


ILLUSTRATION L5500A
SIMPLIFIED BLOCK DIAGRAM OF THE SGI HOST COMPUTER CHASSIS

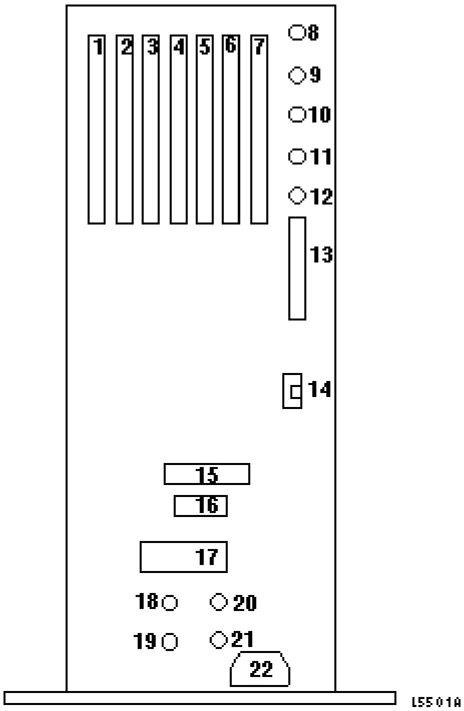
2- OPERATOR WORKSPACE CABINET

2-1 SGI Host Computer

The Computer and Console implement a security system which uses hardware and software to deter the unauthorized use and/or modification of proprietary and purchased option software residing in the Computer. The security system ensures that only authorized users (service person or service contract holder) are allowed to access and execute proprietary software and prohibits execution of purchased option software on unauthorized systems.

The SGI Indigo2 Computer is housed in a chassis that allows for add on boards. Two of the add-on-boards, the EISA and GIO64, face the rear of the chassis and have external access. There is an EMC port cover that covers ports 2 and 3 which are available for optional add-on-devices. All external connections are on the back of the chassis.

The following list identifies the outputs of the host computer subsystem by location (see Illustration L5501A).



1. G1064 Bit3
2. G1064 Open (Slot Blocker in place)
3. G1064 Mardi Gras Video (Slot Blocker in place)
4. G1064 Mardi Gras Board 3
5. G1064 Mardi Gras Board 2
6. G1064 Mardi Gras
7. EISA Serial I/L (Speicalix Card)
8. Audio
9. Audio
10. Audio
11. Audio
12. Audio
13. SCSI
14. 10-BASE T ethernet
15. Interface module power
16. Parallel
17. Ethernet AUI port
18. Serial 1

ILLUSTRATION L5501A
BACK OF INDIGO COMPUTER

2-1-1 Circuit Boards

The Host Computer will utilize a number of Indigo2 workstation circuit boards. The following list of boards are incorporated into the Host Computer.

- 2115457 IP22 Indigo II CPU Motherboard
- 2115457-2 Motherboard Midplane for expansion boards - G1064 and EISA expansion slots
- 2115457-3 MG1,0 - Mardi Gras Graphics Engine - No geometry engine, no texture memory
- 2115457-4 MG1,1 - Mardi Gras Graphics Engine - One geometry engine, 1MB texture memory
- 2115457-5 MG 1,4 - Mardi Gras Graphics Engine - One geometry engine, 4MB texture memory
- 2115457-11 MG VCR - Mardi Gras VCR Option Board
- 2115457-XX IP22 Serial/Keyboard/Mouse/Parallel Port Expansion Board

In addition, the following non-SGI boards are used.

- 2139035PSP Specialix SLXOS serial card with modular terminal adapter (MTA). The MTA will be set up with 4 RS232 and 4 RS422 modules

- 2124215 Bit3 G1064 to VME translator

2-1-2 Circuit Board Details

CPU The Indy II has a RISC 4400 processor with 256 Mega Bytes of RAM. The CT Rhapsody uses the same CPU board with some differences. The CPU is faster (200 Mega Hertz), and has more memory.

It is used for all the host processing, SCAN, Display, and AW functions. The IP or Image Processor functions are performed on the Mardi Gras Graphic Accellerator Boards.

Main Memory The main memory for this system is configured in 72 pin SIMM modules with parity. The minimum memory size is 256 MB and can be upgraded to 512 MB.

Video Graphics Adapters

Input/Output Communications - Built In

Input/Output Communications Adapter(s)

SGI MidPlane GIO64 and EISA Bus

2-1-3 Power Requirements

- AC Power Input
- Main Power Supply
- Auxiliary Power Supply
- External Power Connectors

2-1-4 Internal Disk Devices

This system has the capacity for up to three hard disks (initial release of the product will support one 4 Giga Byte Disk), one floptical disk, an MOD Drive (for archive media) and a CD-ROM drive (used for the software installation.

• **Storage Device Controller(s)** The SGI Host Computer mother board has two built in SCSI-II ports to communicate with peripherals.

• **Hard Disk Drive** The Host Computer can accommodate three hard disk drives with internal access only (there is no operator access). The mounting accepts a standard drive mounted in a sled tray for easy service access. The system interface is a SCSI daisy chain cable with a removable connector which is a 50 position dual row header.

• **Floppy Disk Drive**

• **CD-ROM Drive and MOD Drive**

2-2 IBM Compatible Personal Computer

2-3 Bit 3 Fiber Optic Interface

2-4 Model 800 Ethernet Hub

2-5 Specialix Serial Expansion

2-6 Security

2-7 Laptop Interface

3- OPERATOR WORKSPACE TABLE

3-1 Workspace Interface Module

3-2 Host Monitor

The Operator Workspace Monitor is available in either monochrom (for up to 30 Gauss), or color (for up to 10 Gauss).

The monochrom monitor is 19". It comes with a monitor bezel that accommodates small magnets that will be to used "shim" the image displayed to compensate for magnetic fields nearby.

The color monitor is shielded by micro metal and does active compensation, allowing the color monitor to continue to be functional during the MRI sequence. The bezel used with the color monitor is called an extractable viewing hood.

3-3 LCD Monitor

3-4 Keyboard and Mouse

The Keyboard contains an emergency stop button which can be used by the operator to power down certain system cabinets. When the button is pressed, a signal is sent to the TYME-II Board in the System Cabinet. This sends an E-Stop signal to the PDU, which causes power shut-off to the RF/Pen Cabinet, Gradient Cabinet, Shim Power Supply, and Magnet Power Supply Cabinets.

3-5 Legacy DAT Drive

3-6 Legacy OD Drive

3-7 Power Module

3-8 Workspace Interface Module

3-9 DASM

4- Optional Data Storage Devices - Sheet 1

The following optional external devices can be added to the system:

- Imaging Camera
- Remote Video Monitor
- 1/2" Reel-to-Reel Tape Drive
- Line Printer
- VTR/VCR

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	June 16, 1998	R.Hawthorne	Initial conversion to Word